Summary Report

Volleyball-Group-Activity-Recognition

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Abstract—A condensed summary of the Volleyball Group Activity Recognition project: the data pipeline, the generic multi-baseline loader, and the four completed baselines — B1, a single-frame fine-tuned ResNet-50 (62.6% test accuracy / 0.630 macro F1); B3, a person-then-group crop classifier (60.7% / 0.589); B4, a temporal classifier that runs an LSTM over B1's frozen frame features (66.1% / 0.673, best macro F1); and B5, a per-player temporal model that pools LSTM summaries of each player's crop sequence (66.3% / 0.619, best accuracy). The dataset (Ibrahim et al., CVPR 2016) contains 55 videos / 4,830 clips with 8 scene-level group activities and 9 player-level actions.

Keywords— *Volleyball, Group Activity Recognition, CNN, RNN, LSTM, ResNet50, Classification, Multiclass Classification, Transfer Learning*

1. Problem & Dataset

Volleyball Group Activity Recognition is a hierarchical classification problem: predict the scene-level group activity (8 classes: `l`/`r`-pass, -spike, -set, -winpoint) and the per-player action (9 classes: blocking, digging, falling, jumping, moving, setting, spiking, standing, waiting). The dataset comes from "A Hierarchical Deep Temporal Model for Group Activity Recognition" (Ibrahim et al., CVPR 2016; official repo, Kaggle mirror).

Key facts: **55 videos, 4,830 clips** (train/val/test = 2,152/1,341/1,337 clips over disjoint video splits of 24/15/16). Each clip has **41 frames**; the clip folder is named after its annotated *middle frame*. Both label levels are heavily skewed: `standing` is ≈68% of person actions, and `l/r_winpoint` are ≈2.5× rarer than the other group classes. The group classes come in **left/right mirror pairs**, so horizontal geometry is part of the label — the transform pipelines therefore use no horizontal flips and warp inputs directly to 224×224 so the full court (or the full player) always stays in view.

2. Data Pipeline

Three raw annotation sources (action detections, person detections, player tracking) are unified by a two-stage parser (`src/json_parser.py`) into one master JSON keyed `video_id/clip_id`, then enriched with the group-activity label per clip (looked up per video against each clip's own middle frame) and cached as a pickle (∼247 MB) for instant metadata loading. Frames are served either from a memory-mapped LMDB (local) or straight from disk (Kaggle); both backends expose the identical `VolleyballDataset`, which covers every baseline through flags: full frames or per-player crops, 1 or 9 frames per clip. A custom collate pads the variable player count per clip and returns a validity mask so padded slots never reach the loss or the pooling.

3. Baseline Roadmap

Table 1 lists the planned progression of models, each adding temporal and/or player-level structure on top of the previous one.

Table 1. Baseline models. B1–B7 complete with results (B7 leads); B8 implemented but untrained.

Model	Temporal	Player	Scene
B1	—	—	Image clf.
B3	—	Crop clf.	Concat-pool + MLP
B4	LSTM	—	LSTM
B5	LSTM/player	Concat-pool	MLP
B6	LSTM/image	Max-pool/frame	LSTM + skip Conv1d
B7	$\text{LSTM}_1 + \text{LSTM}_2$	Pool/frame	LSTM_2 + skips
B8	$\text{LSTM}_1 + \text{LSTM}_2$	Team-split	LSTM_2 + skips

4. Baseline 1: Single-Frame Classifier

Fine-tunes a pretrained ResNet-50 on the clip's middle frame for the 8 group classes. Two-stage schedule: linear probe (head only, 10 epochs, lr 10^{-3}), then partial fine-tune of `layer3/4` + head (up to 100 epochs, backbone lr 2.5×10^{-4}, head 3×, cosine annealing, batch 16, frozen BatchNorm held in eval).

Table 2. Baseline 1 test-set metrics.

Metric	Value
Accuracy	62.60%
Macro F1	0.630
Test Loss	1.42

Fig. 1 shows the per-class confusion on the test split.

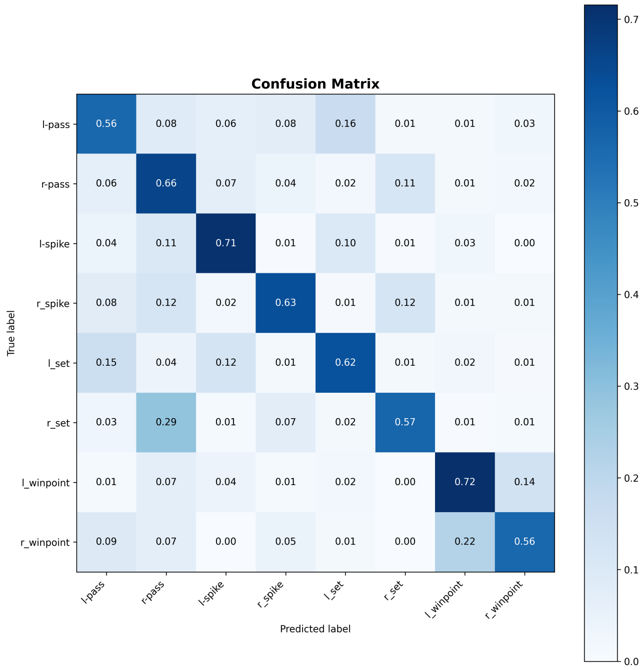

Figure 1. Baseline 1 test confusion matrix.

5. Baseline 3: Person-Then-Group

Stage A fine-tunes ResNet-50 on individual player crops for the 9 person actions with inverse-frequency class-weighted CE (without it the loss collapses onto `standing`). **Stage B** freezes that backbone, extracts a $(P, 2048)$ feature matrix per clip, concatenates masked max- and mean-pooling across players (2×2048), and trains an MLP ($4096 \rightarrow 512 \rightarrow 8$, Dropout 0.4) with class-weighted CE for the group classes. SGD (momentum 0.9, Nesterov), 50 epochs and patience 10 per stage, batch 16 clips, `nn.DataParallel` on multi-GPU hosts. Stage A reaches 70.4% validation accuracy (macro F1 0.512) on the person actions; Stage B 60.0% (0.584) on the group activities.

Fig. 2 shows the dominant error mode: left/right mirror confusion within each activity pair, worst on the rare `r_winpoint`. The main open issue is **overfitting** (train ≈95% vs. val 58–70%), to be addressed with stronger augmentation, higher dropout, and earlier stopping in a follow-up run.

Table 3. Baseline 3 test-set metrics.

Metric	Value
Accuracy	60.73%
Macro F1	0.589
Test Loss	1.09

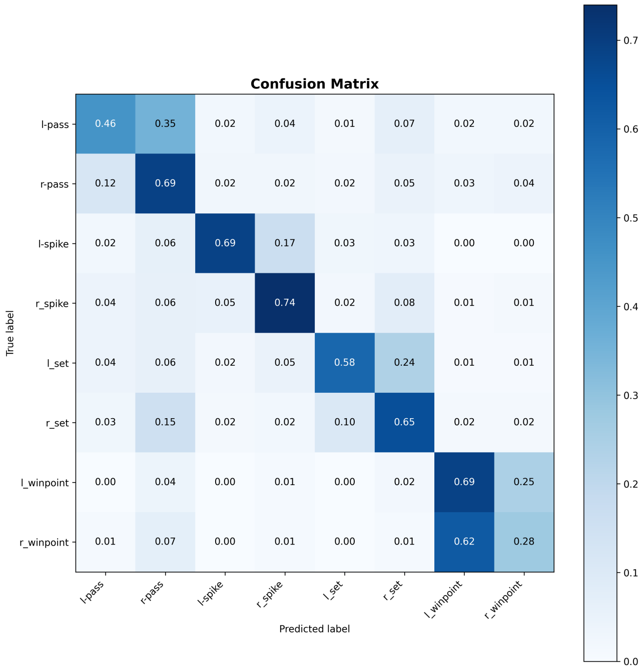

Figure 2. Baseline 3 test confusion matrix.

6. Baseline 4: Temporal Image Classifier

Each clip's 9-frame window is pushed through a **frozen** feature extractor loaded from Baseline 1's fine-tuned backbone, producing a (9, 2048) feature sequence per clip. A single-layer LSTM (hidden 512) consumes the sequence and its final hidden state is classified by a small MLP head (512 → 256 → 8, Dropout 0.3). Only the LSTM + head train (≈5.4M parameters; SGD lr 10^{-3}, cosine annealing, class-weighted CE, batch 16 clips).

Table 4. Baseline 4 test-set metrics.

Metric	Value
Accuracy	**66.12%**
Macro F1	**0.673**
Test Loss	1.05

B4 is the strongest baseline so far: temporal context over B1's own features is worth **+3.5 accuracy points / +0.043 macro F1** versus B1's single-frame result — the first direct evidence in this project that motion carries group-activity signal. Convergence is very fast (best validation F1 at epoch 4; training accuracy ≈99% by epoch 14, where early stopping fired), repeating the overfitting pattern of B1/B3. Fig. 3 shows the test confusion matrix.

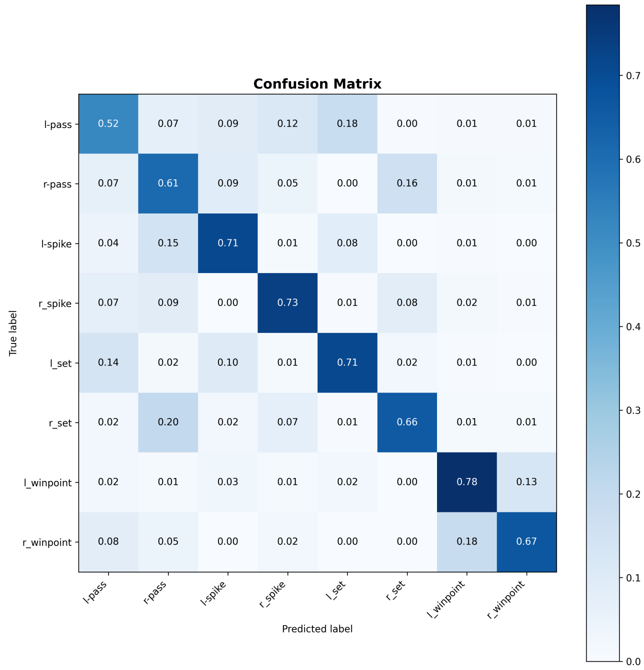

Figure 3. Baseline 4 test confusion matrix.

7. Baseline 5: Temporal Person-Level Model

A two-stage temporal extension of B3. **Stage A** feeds each player's 9-crop sequence through B3's **frozen** Stage-A person-action backbone and trains a shared LSTM (hidden 3072, 1 layer) + head to classify the 9 person actions from the sequence's final hidden state. **Stage B** freezes the Stage-A model entirely, computes one LSTM summary per player per clip, pools the summaries across players with masked concat (max || mean, classifier input 2×3072), and trains an MLP head (6144 → 3072 → 1536 → 8, LayerNorm + Dropout 0.2) on the 8 group activities. Training uses SGD (lr 10^{-3}, cosine annealing in Stage B), class-weighted CE in both stages, and gradient accumulation (effective batch 8 = micro 4×2) since each clip expands to $9 \times \sim 12$ crop images.

Table 5. Baseline 5 test-set metrics.

Metric	Value
Accuracy	**66.34%**
Macro F1	**0.619**
Test Loss	**0.97**

B5 achieves the **best test accuracy and test loss of baselines B1–B5** (66.3% / 0.97 vs. B4's 66.1% / 1.05) and is markedly stronger on the core activities — spike F1 reaches 0.78–0.80 and set 0.71 on both court sides, with cross-activity confusion nearly eliminated — direct evidence that per-player temporal modeling adds signal beyond B4's scene-level features. Its macro F1 (0.619) nevertheless trails B4 (0.673) because of a single failure mode: **the `l/r_winpoint` confusion is the worst left/right confusion of any baseline in this project**. As Fig. 4 shows, `r_winpoint` collapses to 0.08 recall (F1 0.13), with 83% of true `r_winpoint` clips predicted as `l_winpoint`. The cause is structural: orderless max/mean pooling over all ∼12 players erases which side of the net each player occupies; winpoint clips show both teams in near-identical celebration poses, so side identity is the only discriminative signal — and the pooling destroys it. B8's team-split pooling (pool each team's six players separately, then concatenate) directly targets this failure.

8. Baseline 6: Scene-Level Temporal Model

B6 moves the temporal model from the player level (B5) to the **scene level**. Each frame's player crops pass through B3's **frozen** Stage-A backbone and are masked max-pooled across players *per frame*, yielding a (9, 2048) scene sequence per clip. An LSTM (hidden 512, 1 layer) consumes it; instead of keeping only the last hidden state, its 9 hidden states are concatenated *along the time axis* with a linear

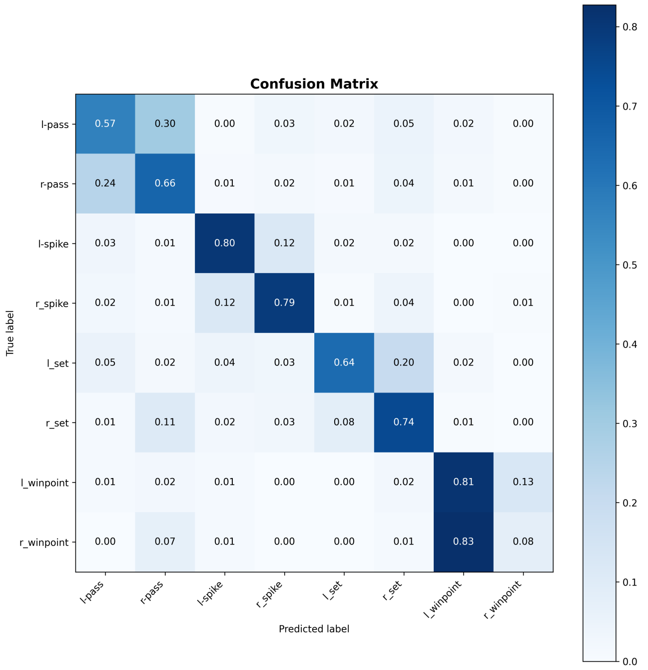

Figure 4. Baseline 5 test confusion matrix. Note the `r_winpoint` row: 83% of its clips are predicted as `l_winpoint` — the side-blind pooling failure discussed in the text.

projection of the pooled per-frame features (skip connection), giving (18, 512), which a two-stage Conv1d with a global temporal kernel (512 → 256, k=18, then 256 → 128, k=1) collapses into a 128-dim clip summary. **Stage A** pretrains LSTM + projection + Conv1d on the 9 person actions using single-player tracks (P=1, pooling is an identity). **Stage B** is two-phase: a 10-epoch **linear probe** (temporal model frozen, MLP head only, lr 10^{-3}), then a **joint fine-tune** that unfreezes the temporal machinery with differential learning rates (temporal 10^{-4}, head 10^{-3}, cosine annealing, 50 epochs). The ResNet extractor stays frozen throughout; training otherwise matches B5 (class-weighted CE, label smoothing 0.01, effective batch 8 = micro 4 × 2).

Table 6. Baseline 6 test-set metrics.

Metric	Value
Accuracy	**70.53%**
Macro F1	**0.686**
Test Loss	**1.04**

B6 is the first baseline to lead on *both* metrics: accuracy gains **+4.2 points over B5** (70.5% vs. 66.3%) and macro F1 passes B4's previous best (0.686 vs. 0.673). Scene-level temporal modeling over pooled person features corresponds to the paper's "two-stage model without LSTM 1" (74.7% there); the remaining ~4-point gap is consistent with our frozen rather than end-to-end backbone. The two-phase Stage B was decisive: the linear probe plateaued at 0.426 validation F1 — a frozen LSTM pretrained only on person actions is a poor group-activity descriptor — while joint fine-tuning lifted it to 0.646 (+0.22). The scene LSTM must see the group labels; the probe merely gives the head a stable start before the temporal weights move.

Per class (Fig. 5), B6 is strongest on the acting-player activities (spike F1 0.86/0.82, set 0.72/0.69) and **halves B5's winpoint collapse without any team-aware pooling**: `r_winpoint` recall recovers from 0.08 to 0.47 (F1 0.13 → 0.49) and `l_winpoint` reaches F1 0.55 — scene-level temporal context evidently carries side information that per-player last-hidden summaries lost. Left/right confusion is nevertheless now the dominant remaining error: 47% of true `r_winpoint` is predicted `l_winpoint`, l/r-pass leak 20–22% into each other, and 21% of `l_set` is called `r_set` — the model is usually right about *what* happened but guesses *which side*. All-player max pooling remains side-blind; B8's team-split pooling targets exactly this. Overfitting persists (Stage A train F1 0.96 vs. val 0.54; fine-tune 0.93 vs. 0.64).

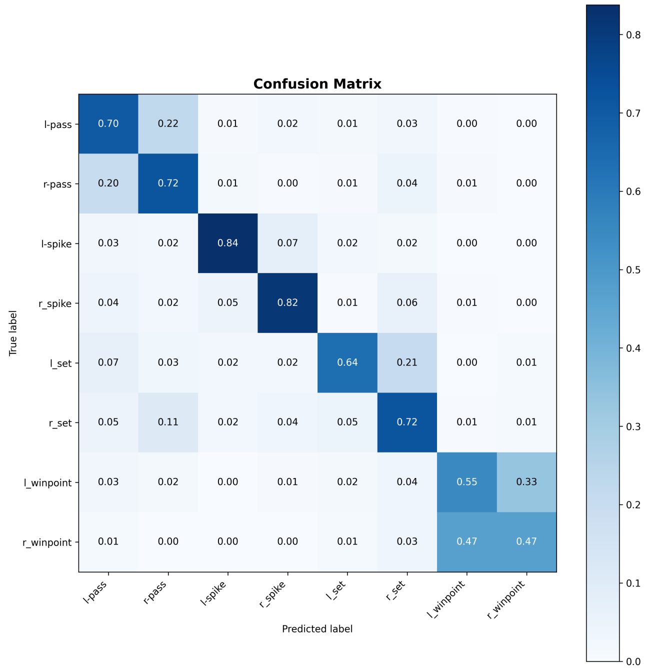

Figure 5. Baseline 6 test confusion matrix. Winpoint recovers from B5's collapse (`r_winpoint` recall 0.08 → 0.47), but left/right leakage — winpoint, pass, and set — is now the dominant error mode.

9. Baseline 7: Full Hierarchical Model

B7 is the paper's two-stage hierarchical model with skip connections at both levels. A player-level **LSTM$_1$** runs over each player's 9-frame track; players are pooled per frame into a scene sequence that a scene-level **LSTM$_2$** consumes. Two skips are added on top of the paper: (1) *player-level, feature-axis* — each timestep's per-player vector is [LSTM$_1$ output || Linear-projected backbone features] (the paper's fc7 || hidden trick), keeping appearance beside the temporal summary while the time axis survives for LSTM$_2$; (2) *scene-level, time-axis* — B6's recipe of concatenating LSTM$_2$'s hidden states with a projection of the pooled scene sequence along time, then a global-kernel Conv1d fusion. Stage A pretrains LSTM$_1$ + projection on the 9 person actions (P=1 tracks); Stage B is two-phase (probe with the player model frozen, then joint fine-tune unfreezing LSTM$_1$), selecting on validation accuracy. Optimizer is AdamW.

Table 7. Baseline 7 test-set metrics (run 3).

Metric	Value
Accuracy	**73.75%**
Macro F1	**0.701**
Test Loss	**0.89**

B7 is the **best baseline in the project**: accuracy +3.2 points over B6 (73.8% vs. 70.5%), macro F1 0.701 vs. 0.686, and the lowest test loss (0.89). The full hierarchy pays off — an adapting player-level LSTM feeding a scene-level LSTM beats B6's single scene-level LSTM over pooled frames. The result hinges on the **two-phase Stage B**: with LSTM$_1$ frozen the probe plateaus at ~0.51 validation accuracy, and unfreezing it for the joint fine-tune lifts validation accuracy to ∼0.68 (+0.17). This is the same lesson as B6 (pretrained temporal weights must adapt to the group task) and explains why B7's earlier *frozen* single-phase run scored only 65.8%, below B6 — unfreezing turned that into +7.9 points and the project lead. What B7 still does not fix is the left/right confusion: it pools all players side-blind, so winpoint/pass/set leak across sides. That is exactly B8's target.

10. Baseline 8: Team-Split Pooling

B8 makes a single architectural change to B7 — the paper's group-style pooling. Instead of pooling all ∼12 players into one scene vector (side-blind), each team's players are pooled *separately* and the two team vectors are concatenated, so the scene vector keeps *which side did what* — the signal every earlier baseline erases, and the direct fix for the dominant left/right confusion (winpoint, pass, set). Both of B7's skips are kept unchanged; only the scene vector doubles in width (LSTM_2 input becomes $4H_1$ for max/mean or $8H_1$ for concat). Team membership comes from the data loader in team mode (`with_teams=True`), which emits a per-player `team_ids` tensor (0/1 by court side, −1 for padding) derived once per clip from box center-x ordering — the paper's split, applied as a fixed per-track label so it coexists with the track-ID ordering the LSTMs require. Uneven or empty teams are handled by masking (a missing team pools to a zero vector).

B8 is implemented and shape-verified (forward/backward for both pool modes, checkpoint round-trip, the zero-player-team edge case, and Stage A ignoring `team_ids` all pass a synthetic test) but **not yet trained**. It is the ablation rung that isolates team structure: in the extension paper, team-split pooling is worth +11.6 **points (**70.3 → 81.9**)** over all-player pooling, and it targets exactly the left/right errors the confusion matrices have shown since B5. One caveat: quality hinges on the loader's x-coordinate team assignment, which can mislabel players near mid-court — if B8 does not beat B7, inspect the split before the pooling.

11. Next Steps

(i) **Train and evaluate B8** — its team-split pooling (already implemented and shape-verified) is the direct fix for the left/right winpoint/pass/set confusion that persists through B7, and the paper's largest single ablation gain (+11.6 points there); built on B7's 73.8% it should push higher if the x-coordinate team assignment is clean. (ii) Reduce overfitting across all baselines — stronger augmentation, higher dropout, and earlier stopping remain the levers (B7's fine-tune still reaches ∼94% train accuracy against ∼68% validation).